Global Technologies

XD12

The LED Current Sink Driver IC

12 Channels

General Description

The XD12 is a 12-channel constant current sink driver for mini-LED Backlight display applications in AM direct driving methodology. It supports max 64mA and 40V at each channel output pin rating. This device can support max 14-bit PWM dimming resolution and a 12-bit DAC global brightness resolution.
The self-protections features, LED open/short detection and thermal shutdown, are embedded. In addition, a Feedback control for adjusting a VLED power is included.
It is used with a XC24 controller as a companion chip. A lot of the XD12 can be daisied to a XC24 channels and controlled through a serial interface by a XC24 controller.

Application

LED back-lighting with local dimming

Features

- Bi-directional 1-wire interface and configurable transmit rate
- Daisied connection between devices and pin layout suitable for single layer PCB
- 12 independent channels in a package
- Maximum 40V headroom voltage
- Configurable max current level (4/8/12/16/24/32/46/64mA)
- 12-bit DAC global brightness control
- 14-bit PWM dimming
- Configurable dimming directions, Head shift lighting and Tail shift lighting
- Sync controllability between daisied devices
- Sink current accuracy ±1% (@ configured max current * 1)
- Configurable LED SHORT and OPEN detection level
- Configurable FEEDBACK level
- Configurable Over-Temperature shutdown @ Typical 135℃
- Configurable Over/Under-Voltage of internal LDO
- Supply voltage 4.5 ~ 5.5V
- 24-PIN QFN 3030 package
- Latch-up Immunity, I_{LATCH} (JEDEC JESD78D Nov 2011) : ±100mA
- ESD protection, ESD_{HBM} (Norm: MIL 883 E Method 3015 Human Body Model) : ±2KV

Block Diagram

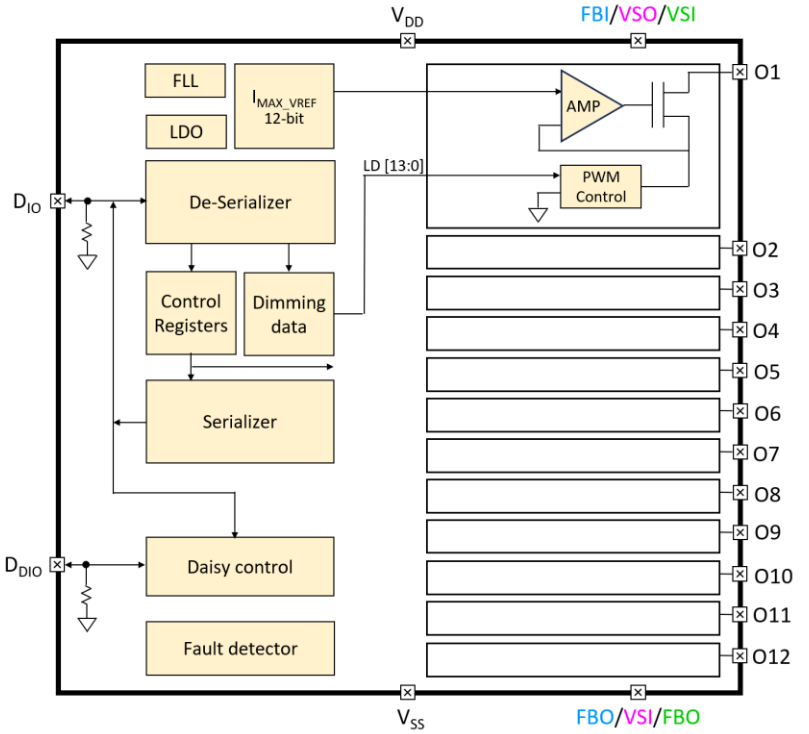


Figure 1. Block Diagram

Application Circuit

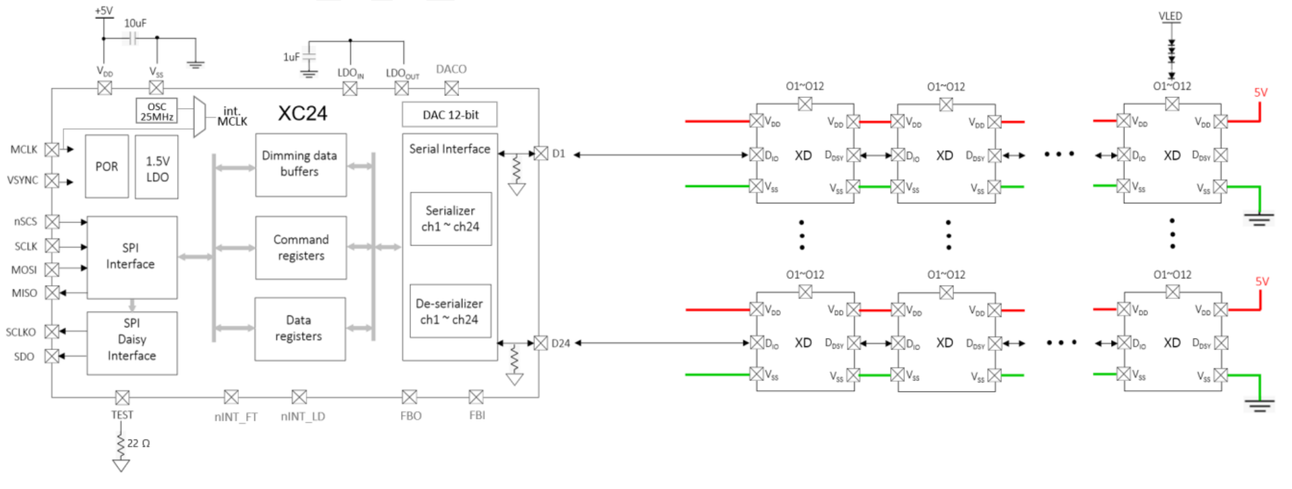


Figure 2. Application scheme

Pin diagram (Top View)

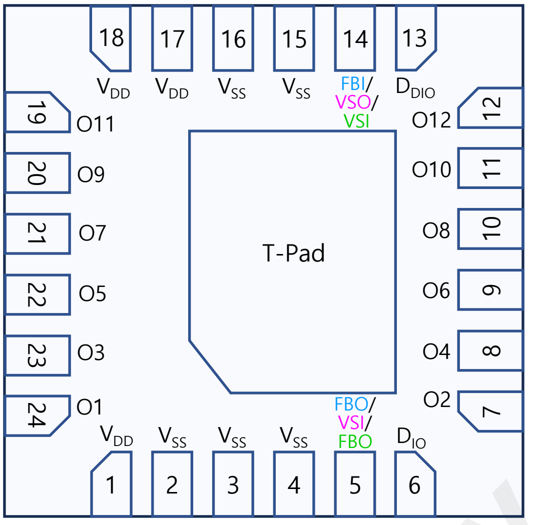


Figure 3. Pin diagram

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
V_{DD}	1	-	Power	Supply voltage	
V_{SS}	2	-	GND	Ground	
V_{SS}	3	-	GND	Ground	
V_{SS}	4	-	GND	Ground	
FBO/VSI/FBO [1)]	5	I/O	Digital	FBO/ VSI/ FBO (Configurable through IO_MODE)	FBO is Open-drain output If not used, connect to GND
D_{IO}	6	I/O	Digital	Data In/Out	Internal PD, 48KΩ
O2	7	O	Current Load	Current Sink 2	If unused, connect to GND
O4	8	O	Current Load	Current Sink 4	If unused, connect to GND
O6	9	O	Current Load	Current Sink 6	If unused, connect to GND
O8	10	O	Current Load	Current Sink 8	If unused, connect to GND
O10	11	O	Current Load	Current Sink 10	If unused, connect to GND
O12	12	O	Current Load	Current Sink 12	If unused, connect to GND
D_{DIO}	13	I/O	Digital	Daisied Data In/Out	Internal PD, 48KΩ
FBI/VSO/VSI [1)]	14	I/O	Digital	FBI/ VSO/ VSI (Configurable through IO_MODE)	If not used, connect to GND

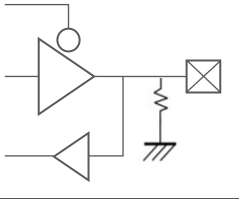
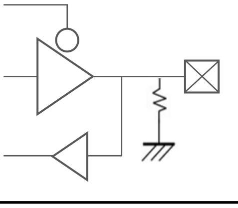
Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
V_{SS}	15	-	GND	Ground	
V_{SS}	16	-	GND	Ground	
V_{DD}	17	-	Power	Supply Voltage	
V_{DD}	18	-	Power	Supply Voltage	
O11	19	O	Current Load	Current Sink 11	If unused, connect to GND
O9	20	O	Current Load	Current Sink 9	If unused, connect to GND
O7	21	O	Current Load	Current Sink 7	If unused, connect to GND
O5	22	O	Current Load	Current Sink 5	If unused, connect to GND
O3	23	O	Current Load	Current Sink 3	If unused, connect to GND
O1	24	O	Current Load	Current Sink 1	If unused, connect to GND

Note : 1) The pin function can be configured through the IO_MODE register bits. If the pin will be used as the FBO, then the pin acts as an Open-drain output. (external PU resistor should be attached)

Table 1. Pin Description

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
D_{IO}	I/O		Data In/Out
D_{DIO}	I/O		Daisied Data In/Out

Equivalent Circuits of individual pins

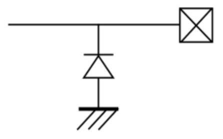
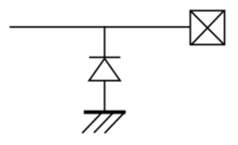
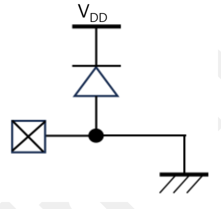
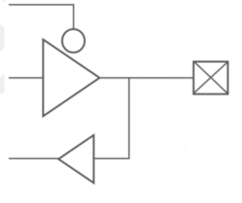
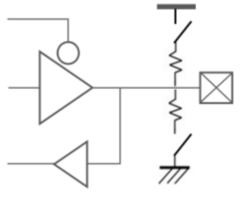
Pad Name	I/O	Internal Circuit	Description
O1 ~ O12	Current Sink		Current Load
VDD	-		Supply voltage
VSS	-		GND
FBO/VSI/FBO	I/O		1. FB (daisied) Output (OD) 2. VSYNC (daisied) Input 3. FB Output (Open-drain) ※ if not used, connect to GND
FBI/VSO/VSI	I/O		1. FB (daisied) Input 2. VSYNC (daisied) Output 3. VSYNC Input ※ if not used, connect to GND

Figure. 4 Equivalent Circuits of individual pins

Maximum Absolute ratings (Ta = 25℃ unless otherwise noted)

Parameter	Symbol	Value	Unit
Supply voltage	V_{DD}	-0.3 ~ 6.5	V
Sink output pin voltage	V_O	40	V
Sink output current	I_O	64	mA
D_{IO} to GND	V_{DIO}	-0.3 ~ V_{DD}+0.3	V
D_{DIO} to GND	V_{DDIO}	-0.3 ~ V_{DD}+0.3	V
Thermal Resistance			
Junction to Ambient	$R_{\theta JA}$	TBD [1)]	℃/W
Junction to Case	$R_{\theta JC}$	TBD [1)]	℃/W
Power Dissipation on package	P_{TOT}	TBD [1)]	W
Junction temperature	T_J	-20 ~ 150	℃
Storage temperature	T_{STG}	-45 ~ 150	℃
Relative Humidity (non-condensing)	RH_{NC}	5 ~ 85	%
Moisture Sensitivity	MSL	Maximum Floor Life-time	3

Stresses beyond the Maximum Absolute ratings may cause permanent damage to the device
Note : 1) Surface mounted on FR-4 substrate PC Board 1” x 1” inch 2-oz copper plate

Table 2. Maximum Absolute ratings

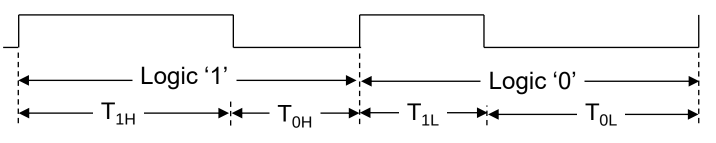
Electrostatic Discharge

Parameter	Symbol	Value	Unit	Remark
Electrostatic Discharge HBM ; Contact	ESD_{HBM}	±2K	V	C=100 pF, R=1.5 kΩ
Electrostatic Discharge CDM ; Contact	ESD_{CDM}	±500	V	R=1 Ω

Table 3. Electrostatic Discharge

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0℃ unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Supply voltage						
Operating voltage	V_{DD} 1)		4.5	5.0	5.5	V
Supply current						
Quiescent Current	I_S	Standby state, No LED current	-	-	TBD	mA
Operating Current 1	I_{DD1}	All output sink current = 4mA	-	-	TBD	mA
Operating Current 2	I_{DD2}	All output sink current = 8mA	-	-	TBD	mA
Operating Current 3	I_{DD3}	All output sink current = 12mA	-	-	TBD	mA
Operating Current 4	I_{DD4}	All output sink current = 16mA	-	-	TBD	mA
Operating Current 5	I_{DD5}	All output sink current = 24mA	-	-	TBD	mA
Operating Current 6	I_{DD6}	All output sink current = 32mA	-	-	TBD	mA
Operating Current 7	I_{DD7}	All output sink current = 46mA	-	-	TBD	mA
Operating Current 8	I_{DD8}	All output sink current = 64mA	-	-	TBD	mA
Oscillator						
Internal MCLK	f_{OSC}			40		MHz
In/Out						
D_{IO}, D_{DIO}	V_{IL}	Input logic low level	-	-	0.8	V
	V_{IH}	Input logic high level	2.7	-	V_{DD}	V
	V_{OL}	Output logic low level	-	-	0.8	V
	V_{OH}	Output logic high level	V_{DD} * 0.8	-	V_{DD}	V
	R_{PD}	Internal pull-down resistance	-	48	-	KΩ
Logic high time 2)	T_{1H}, T_{0L}	as a receiver	6	-	63	MCLK
Logic low time 2)	T_{1L}, T_{0H}	as a receiver	3	-	32	MCLK
Propagation delay	T_{D_DS}	Downstream (D_{IO} to D_{DIO})	-	9.0	14	ns
Propagation output skew3)	T_{DSKEW_DS}	D_{DDIO} T_{1H}=T_{1H}-T_{DSKEW_DS} @downstrm	-	1.5	2.6	ns
Propagation delay	T_{D_US}	Upstream (D_{DIO} to D_{IO})	-	9.5	14.5	ns
Propagation output skew3)	T_{DSKEW_US}	D_{DIO} T_{1H}=T_{1H}-T_{DSKEW_US} @upstream	-	1.4	2.3	ns



Note : 1) The operation below V_{DD}, 5V will limit a maximum LED sinking current and affect a current variation.
2) The conditions, ①T_{1H} ≥ T_{0H}* 2 ② T_{0L} ≥ T_{1L}* 2 ③ (T_{1H} +T_{0H}) = (T_{0L}+T_{1L}) ≤ 95, are recommended.
3) On downstreaming (or upstreaming), the T_{1H} of D_{DDIO} (or T_{1H} of D_{DIO}) is shorted as many as T_{DSKEW_DS} (or T_{DSKEW_US}), but the total of 1 bit cycle will remain the same. *So, the proper value for T_{1H} and T_{0L} must be carefully determined based on the daisy size.*

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0℃ unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Output						
Driving voltage	V_O	Headroom voltage V_{HD} : 0.18V@ I_O 4mA, 0.21V@ I_O 8mA 0.24V@ I_O 12mA, 0.27V@ I_O 16mA 0.32V@ I_O 24mA, 0.38V@ I_O 32mA 0.47V@ I_O 46mA, 0.60V@ I_O 64mA	V_{HD}	-	40	V
Short detection voltage 4)	V_{SD}	Short detection level V_{SD} : 3b'000=3V, 3b'001=4V 3b'010=6V, 3b'011=8V 3b'100=12V, 3b'101=16V 3b'110=24V, 3b'111=36V	V_{SD}	-	-	V
Open detection voltage 4)	V_{OD}		-	-	0.25	V
Sink current	I_O	Device max current level : I_O 3'b000 = 4mA, 3'b001 = 8mA 3'b010 = 12mA, 3'b011 = 16mA 3'b100 = 24mA, 3'b101 = 32mA 3'b110 = 46mA, 3'b111 = 64mA	-	-	64	mA
Current accuracy 5)	%	@ I_O =Imax(32mA)*0.03	-2.0	-	+2.0	%
		@ I_O =Imax(32mA)*0.1	-1.5	-	+1.5	%
		@ I_O = Imax(32mA)*1	-1	-	+1	%
LED off current	I_{OFF}	Vo = 40V	-	-	300	nA
Feedback						
FB threshold voltage (Logically ORed for 12-ch)	V_{FBA}	FB level V_{FBA} : 3'b000 = 0.40V, 3'b001 = 0.50V 3'b010 = 0.60V, 3'b011 = 0.70V 3'b100 = 0.85V, 3'b101 = 1.00V 3'b110 = 1.15V, 3'b111 = 1.30V	-	V_{FBA}	-	V
FB hysteresis voltage	V_{FBH}		40	60	110	mV
Thermal protection						
Thermal cut off @T_J	T_{CUT}	OFS_TEMP [3:0] = 0x8	-	135	-	℃
Thermal cut off hysteresis @T_J	T_{HYS}		10	20	30	℃

Note : 4) If LED open or short event is happened, the corresponding channel will be turned-off under the function enabled. If the thermal is happened, all channels output will be turned-off under the function enabled.
5) Under satisfying the power dissipation specification.

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0℃ unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
V_{DD} Ramp up and V_{LED} Start						
V_{DD} Ramp up time	T_{VDD}	from 0 to stable 5V	-	-	1.2	ms
V_{LED} Start time	T_{SU_DIM}	after T_{SU_DIM} is completed	6)	-	-	ms
Initialization after V_{DD}, 5V stable						
Internal Initial time	T_{INIT}	from stable 5V	10	-	-	ms
Allowed 1st command	T_{FCMD}	after T_{INIT} is completed	5	-	-	ms
Timeout : Internal timer starts after completing to receive a read-back command and stops after first response bit from a last daisied device. This depends on the frequency of MCLK.						
Internal Timeout 7)	$T_{TIMEOUT}$		-	32,767	-	MCLK
VSYNC daisy out @ EXT_VS_E = 1						
VSYNCO delay	T_{VSOD}	from VSYNCI to VSYNCO	-	5.0	9.0	ns

Note : 6) The time depends on the system configuration, the daisied XD number and Baud-rate.
7) If the timeout will be expired, the internal communication logic will go to an initial state, so new command transaction will be possible. The timeout function is configurable.

Table 4. Electrical Characteristic

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

Power Sequence

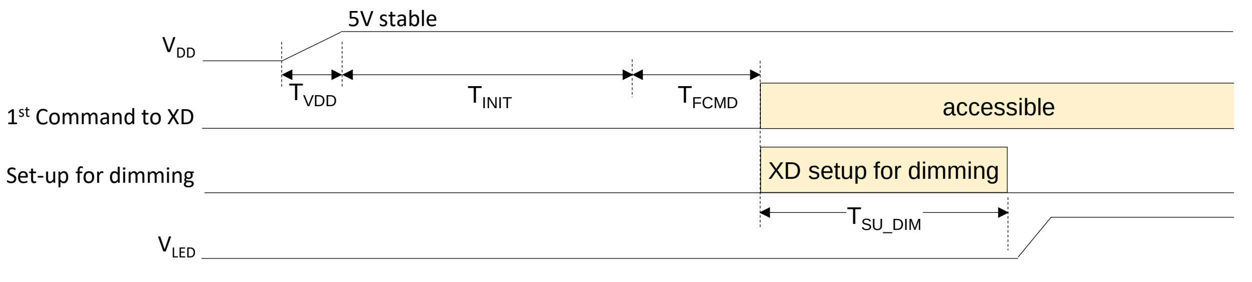
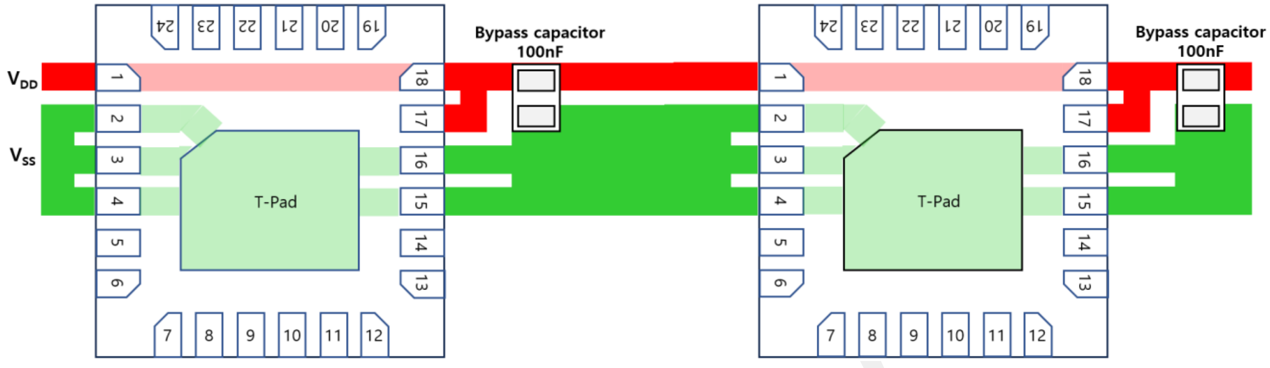


Figure 5. Power up sequence and Ramp-up time

Layout for Power Lane for single-layer PCB

As shown in the figure below, it is recommended that a bypass capacitor should be placed as close to the pin 18 and 17 as possible in below.



Functional Description

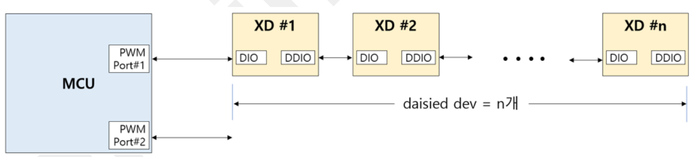
Serial Communication Protocol

The XD can be controlled with several commands through a XC controller or directly by a Micom. The access protocol and frame format are as follows and if several XD is connected in daisy, the commands should be repeated successively as many as the XD number in daisy chain.

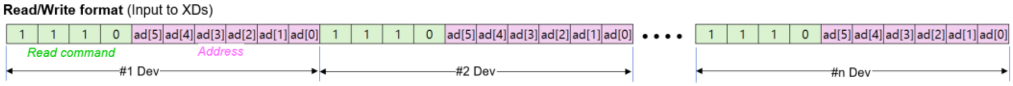
PROTOCOL	frame format																					
write command	bit21	bit20	bit19	bit18	bit17	bit16	bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
1101	start	cmd code [2:0]			addr [5:0]						data [11:0]											
read command	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0												
1110	start	cmd code [2:0]			addr [5:0]																	
readout response	bit20	bit19	bit18	bit17	bit16	bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
1110	start	cmd code [2:0]			Id [4:0]					data [11:0]												
Id transmit	bit19	bit18	bit17	bit16	bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0		
1111	start	cmd code [2:0]			Id [15:0]																	
fault read command	bit8	bit7	bit6	bit5																		
1010	start	cmd code [2:0]																				
fault readout response	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0														
1010	start	cmd code [2:0]			thermal	short	open	fb														
sync gen command	bit4	bit3	bit2	bit1																		
1001	start	cmd code [2:0]																				
id gen	bit4	bit3	bit2	bit1																		
1000	start	cmd code [2:0]																				

Table 5. Serial communication protocol

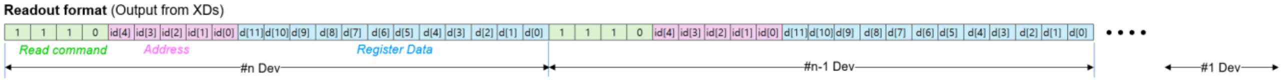
The serial communication protocol and the packet formats are shown in below for more details.



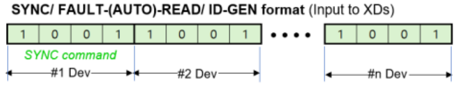
- **Read/Write format** (Input to XDs)



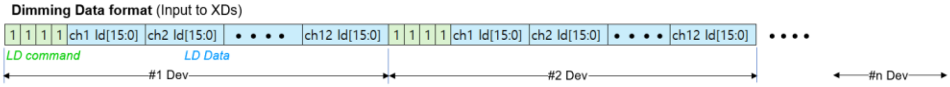
- **Readout format** (Output from XDs)



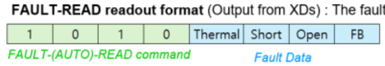
- **SYNC/ FAULT-(AUTO)-READ/ ID-GEN format** (Input to XDs)



- **Dimming Data format** (Input to XDs)



- **FAULT-READ readout format** (Output from XDs) : The fault data, 4 bits, are the data that all daisied XD devices' each fault is ORed.



Functional Description

PWM Configuration

The PWM behavior depends on the PWM configuration set to the PWM_RES and SCAN_NO [2:0] as shown in the Table 6.

x don't care   valid bit

PWM_RES	SCAN_NO [2:0]	sub-frame #	LD Data [15:0]																step # /sub-frame	pwm freq hz@120
			D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0		
0 (12-bit)	0	1	x	x	x	x	11	10	9	8	7	6	5	4	3	2	1	0	4,095	120
	1	2	x	x	x	x	x	10	9	8	7	6	5	4	3	2	1	0	2,047	240
	2	4	x	x	x	x	x	x	9	8	7	6	5	4	3	2	1	0	1,023	480
	3	8	x	x	x	x	x	x	x	8	7	6	5	4	3	2	1	0	511	960
	4	16	x	x	x	x	x	x	x	x	7	6	5	4	3	2	1	0	255	1,920
	5	32	x	x	x	x	x	x	x	x	x	6	5	4	3	2	1	0	127	3,840
1 (14-bit)	0	1	x	x	13	12	11	10	9	8	7	6	5	4	3	2	1	0	16,383	120
	1	2	x	x	x	12	11	10	9	8	7	6	5	4	3	2	1	0	8,191	240
	2	4	x	x	x	x	11	10	9	8	7	6	5	4	3	2	1	0	4,095	480
	3	8	x	x	x	x	x	10	9	8	7	6	5	4	3	2	1	0	2,047	960
	4	16	x	x	x	x	x	x	9	8	7	6	5	4	3	2	1	0	1,023	1,920
	5	32	x	x	x	x	x	x	x	8	7	6	5	4	3	2	1	0	511	3,840

Table 6. PWM Configuration

Functional Description

Local dimming

The Local dimming sequences according to the PWM configuration are shown as follows below, and has two dimming out directions, Head Shift Lighting and Tail Shift Lighting. The set of PWM frequency (pwmclk) according to the Frame frequency can be shown in the Table 7.

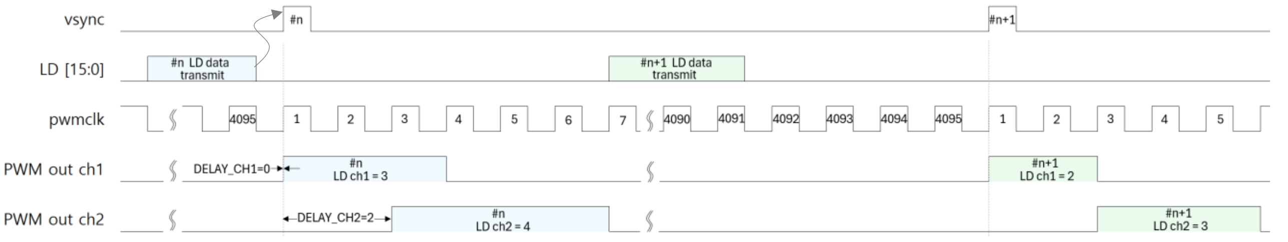


Figure. 6 Head Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=0

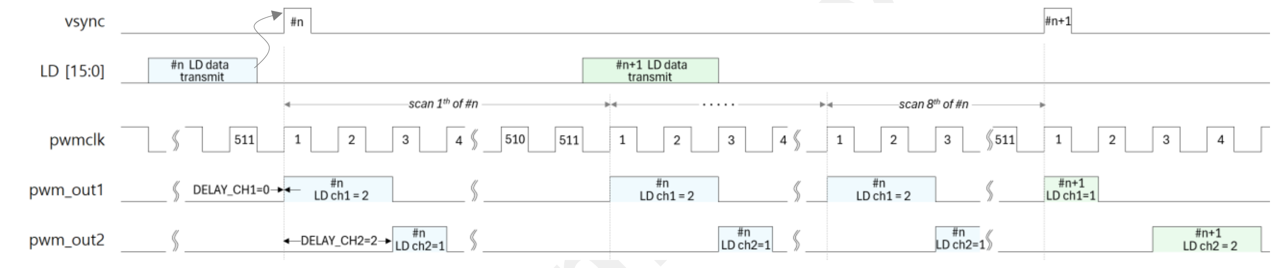


Figure. 7 Head Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=3

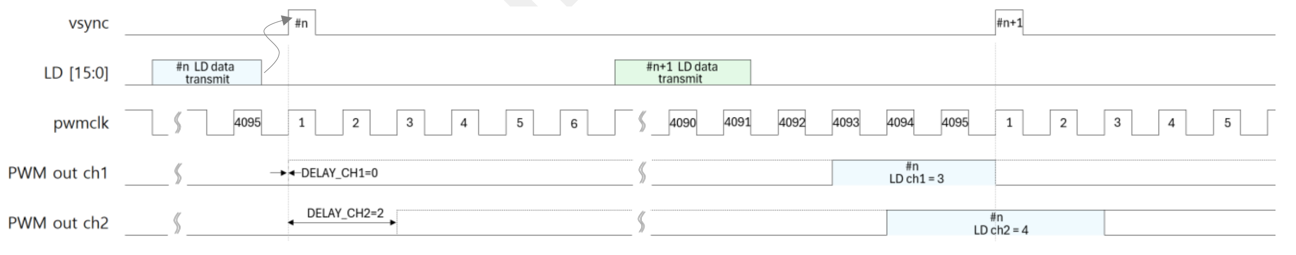


Figure. 8 Tail Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=0

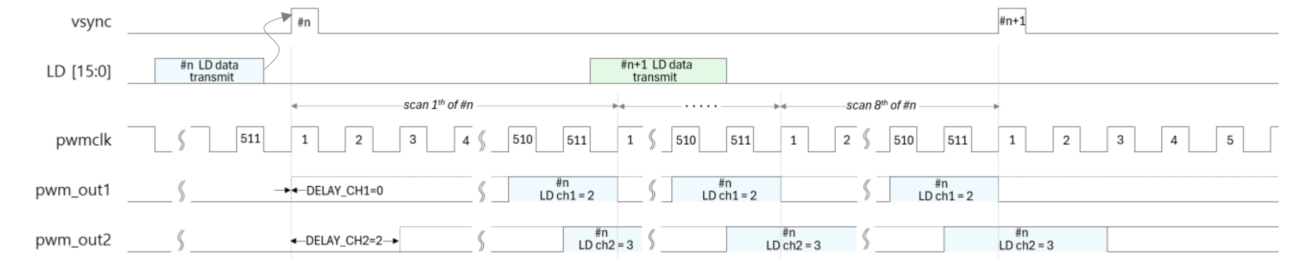


Figure. 9 Tail Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=3

Functional Description

Optional external VSYNC and FB

The XD has 3 external signal IO modes which can be configured through the IO_MODE bits in the LD CONTROL register. This function of each mode and the connection scheme is shown as follows.

IO_MODE [1:0]	Description	FBI	FBO	VSI	VSO	Remark
00	NOP (default)	-	-	-	-	The pins, 5 and 14, can be connected to GND. The FB and VSYNC signal can be replaced by the communication method.
01	VSYNC daisy mode	-	-	used	used	The FB signal can be replaced by the communication method.
10	FB daisy mode	used	used	-	-	The VSYNC signal can be replaced by the communication method.
11	VSYNC and FB non-daisy mode	-	used	-	used	The VSYNC and FB signal can be replaced by the communication method.

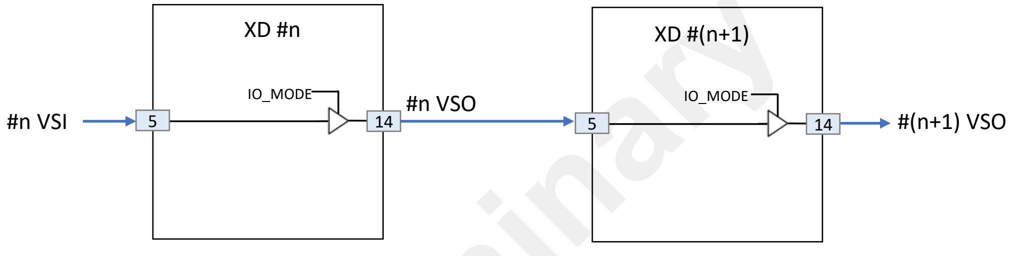


Figure 10. External VSYNC In and Out Scheme

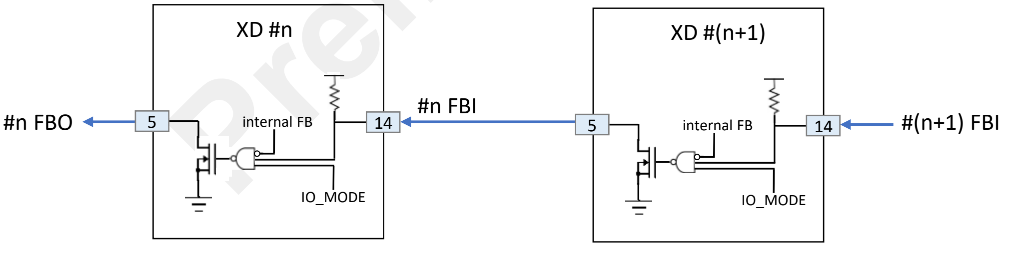


Figure 11. External FBI and FBO Scheme

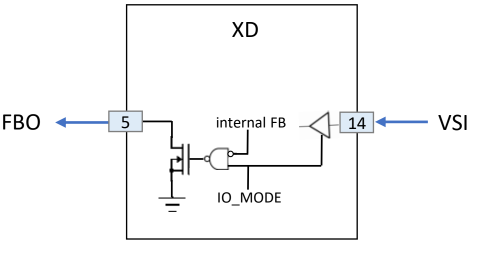


Figure 12. External FBO and VSI Scheme

Functional Description

Switching a frame frequency without a flicker

transaction#	times vsync	ready	3	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	3	3	3	3	3	3	3	3	finish
frame	[hz]	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	118.75	117.50	116.25	115.00	113.75	112.50	111.25	110.00	108.75	107.50	106.25	105.00	103.75	102.50	101.25	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
writing to XD	**addr**																																				
fpwm_div	0x02	79	79	79	79	79	79	79	79	79	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	95	95	95	95	95	95	95	95	95	95
mclk_lock_cnt	0x27/28	328910	329238	329566	329893	330221	330548	330876	331204	331646	331646	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397,975	397975	397444	397051	396658	396265	395872	395479	395086	394692
mclk_lock_cnt_e	0x28	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0

transaction#	time vsync	normal	3	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	3	3	3	3	3	3	3	3	finish
frame	[hz]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	101.25	102.50	103.75	105.00	106.25	107.50	108.75	110.00	111.25	112.50	113.75	115.00	116.25	117.50	118.75	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
writing to XD	**addr**																																				
fpwm_div	0x02	95	95	95	95	95	95	95	95	95	95	94	93	92	91	90	89	88	87	86	85	84	83	82	81	80	79	79	79	79	79	79	79	79	79	79	79
mclk_lock_cnt	0x27/28	394692	395086	395479	395872	396265	396658	397051	397444	397975	397975	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331,646	331646	331204	330876	330548	330221	329893	329566	329238	328910
mclk_lock_cnt_e	0x28	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0

Figure 13. PWM 12-bit

transaction#	time vsync	ready	3	3	3	3	3	3	3	3	3	1	1	1	1	1	3	3	3	3	3	3	3	3	finish
frame	[hz]	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	115.00	110.00	105.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
writing to XD	**addr**																								
fpwm_div	0x02	19	19	19	19	19	19	19	19	19	19	20	21	22	23	23	23	23	23	23	23	23	23	23	23
mclk_lock_cnt	0x27/28	328971	329298	329626	329954	330281	330609	330937	331264	331707	331707	398,048	398,048	398,048	398,048	398,048	398048	397517	397124	396731	396338	395944	395551	395158	394765
mclk_lock_cnt_e	0x28	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0

transaction#	time vsync	normal	3	3	3	3	3	3	3	3	3	1	1	1	1	1	3	3	3	3	3	3	3	3	finish
frame	[hz]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	105.00	110.00	115.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
writing to XD	**addr**																								
fpwm_div	0x02	23	23	23	23	23	23	23	23	23	23	22	21	20	19	19	19	19	19	19	19	19	19	19	19
mclk_lock_cnt	0x27/28	394765	395158	395551	395944	396338	396731	397124	397517	398048	398048	331,707	331,707	331,707	331,707	331,707	331707	331264	330937	330609	330281	329954	329626	329298	328971
mclk_lock_cnt_e	0x28	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0

Figure 14. PWM 14-bit

Register Map

RESET / ID

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x00	0x000	RST	x	x	x	x	x	x	ID [4:0]				

Bits	Fields		Descriptions
0	RST	0	NOP
		1	All internal logics are initialized to a default state and this bit will be auto-cleared after internal initialization.
4:0	ID		These bits are read only. If the XD12 will receive the ID-Gen command from the XC24 controller, then the ID value is automatically generated. The ID value is needed for proper reading, so, the ID-Gen command must be firstly executed by the XC24 controller before executing any commands. If all XD12 drivers daisied will receive the ID-Gen command, the last-located device will have the first number of ID, and the first located device will have the last number of ID, that is, the number of daisied devices. The ID value can not be written by a Host.

LD CONTROL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x01	0x000	CH_SIZE [3:0]				IO_MODE [1:0]		SCAN_NO [2:0]			OVER_TO_E	PWM_RES	LD_DIR

Bits	Fields	Descriptions
		This register determines the dimming sequence. Even if the LD_DIR, PWM_RES, OVER_TO_E and SCAN_NO value will be changed in the middle of the current frame, the value will affect the dimming sequence from the next frame after completing the current frame process.
11:8	CH_SIZE	This value determines how many LD data will be transmitted for one time. This value is not needed to match with the number of enabled channels in the CHANNEL ENABLE register. The LD data to be transmitted will be mapped sequentially from channel 1 to channel n regardless of the corresponded CHx_EN bit activation.
7:6	IO_MODE	These bits determine the FB and VSYNC functional usage as a pin. Refer to below table.

Value	Description	FBI	FBO	VSI	VSO	Remark
00	NOP (default)	-	-	-	-	VSYNC and FB can be got through a communication. The pins, 5 and 14, can be connected to GND.
01	VSYNC daisy	-	-	used	used	External VSYNC signal scheme in a daisy FB can be got through a communication.
10	FB daisy	used	used	-	-	External FB signal scheme in a daisy VSYNC can be got through a communication.
11	VSYNC and FB non-daisy	-	used	-	used	External VSYNC and FB signal scheme VSYNC and FB can be got through a communication.

Bits	Fields	Descriptions
5:3	SCAN_NO	This value determines the number of scanning within a frame. The scanning number is one if the value is zero. In case the value is greater than 0, the larger the value, the smaller the PWM depth. The available value is from 0 to 5, and the relationship between the set value vs the scanning number and the PWM depth can be showed in below Table 6.
2	OVER_TO_E	Over Turn-On Enable. This bit is an optional and recommended to decide the set according to the dimming performance from internal PWM clock. It is useful in case the PWM clock is faster than ideal frequency under full PWM data.
1	PWM_RES	This bit determines the PWM resolution. 0 12-bit PWM resolution (max steps 4,095) 1 14-bit PWM resolution (max steps 16,383)
0	LD_DIR	This bit determines the direction of modulated edge with increasing on time from a vsync. 0 Head Shift Lighting 1 Tail Shift Lighting

FPWM DIVIDER

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x02	0x000	x	x	x	x	x	FPWM_DIV [6:0]						

Bits	Fields	Descriptions
6:0	FPWM_DIV	This register determines the PWM reference clock. The PWM reference clock can be created by dividing the MCLK frequency by the setting value + 1. Even if the setting value will be changed in the middle of the frame, the value will be applied immediately from the next vsync assertion. The setting value according to the frame frequency is guided as shown Table 7 below.

PWM_RES	Register	100Hz	120Hz	144Hz	165Hz	240Hz
0	FPWM_DIV	95	79	67	59	39
	MCLK_LOCK_CNT	393,120	327,600	278,460	245,700	163,800
1	FPWM_DIV	23	19	16	14	9
	MCLK_LOCK_CNT	393,192	327,660	278,511	245,745	163,830

Table 7. FPWM Configuration

CHANNEL ENABLE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x03	0x000	CH12_EN	CH11_EN	CH10_EN	CH9_EN	CH8_EN	CH7_EN	CH6_EN	CH5_EN	CH4_EN	CH3_EN	CH2_EN	CH1_EN

Bits	Fields	Descriptions
11:0	CHx_EN	The CHx_EN determines which channel will be activated. The set value is independent of the value of the CH_SIZE. If the CHx_EN is not set, then the corresponding channel sink output including the its faults, OPENx, SHORTx, FBx, will be deactivated. The CHx_EN bits *should be set consecutively*, such as CH1_EN, CH2_EN, CH3_EN, and so on, and the number of enabled channels does not need to match the CH_SIZE value.

FAULT STATUS

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
RO	0x05	0x000	MISS_VS	x	OV15	UV15	x	x	x	x	THERMAL	SHORT	OPEN	FB

Bits	Fields	Descriptions
11	MISS_VS	The MISS_VS bit indicates whether the vsync missing event happens, and auto-cleared after readout this register.
9	OV15	The OV15 indicates whether the internal LDO keeps over 1.8V for 100us or not, and auto-cleared after a readout this register.
8	UV15	The UV15 indicates whether the internal LDO keeps under 1.2V for 100us or not, and auto-cleared after a readout this register.
3	THERMAL	The THERMAL bit indicates whether the THERMAL event due to overheating happens, and this bit will keep asserted as long as the THERMAL event is not cleared. The event threshold can be set by the OFS_TEMP in the TEMP register. If the T_OFF_E bit of the FAULT CONTROL register is set, all output sink drivers will be turned-off as long as the event is asserted, and turned-on again after internal temperature goes down below the set level of the OFS_TEMP. This bit involves other fault events in the readout information, MISS_VS, OV15 and UV15 at the FAULT_(AUTO)_READ transaction.
2	SHORT	The SHORT bit indicates whether the SHORT event happens. The event can be detected in case some channel's headroom goes above the set value in the SHORT_LEVEL of FAULT LEVEL register and its channel's PWM out is ON status. If the S_OFF_E bit of the FAULT CONTROL register is set, the output of the channels in which the event occurs is turned off. This bit is auto-cleared after readout but set again as long as the event is not cleared.
1	OPEN	The OPEN bit indicates whether the OPEN event happens. The event can be detected in case some channel's headroom goes below 0.25V and its channel's PWM out is OFF status. If the O_OFF_E bit of the FAULT CONTROL register is set, the output of the channels in which the event occurs is turned off. This bit is auto-cleared after readout but set again as long as the event is not cleared.
0	FB	The FB bit indicates whether the output channels' headroom voltage is below the value set by FB_LEVEL of the FAULT LEVEL register due to insufficient VLED. This bit is set whenever the FB event occurs in even one of the activated channels and will be auto-cleared after readout this register. This bit will be set again after readout as long as the FB condition is still valid. If the O_FB_E of FAULT CONTROL register is set, the FB event will happen in case the OPEN event is not asserted.

FAULT LEVEL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x06	0x000	x	DEV_MAX_CURR_LEVEL [2:0]			x	SHORT_LEVEL [2:0]			x	FB_LEVEL [2:0]		

Bits	Fields	Descriptions
10:8	DEV_MAX_CURR_LEVEL	The DEV_MAX_CURR_LEVEL determines the physical maximum current level of the device. The max dimming sink-current set by the MAX_CURR_VREF can not be beyond the DEV_MAX_CURR_LEVEL. 3b'000 : max 4mA 3b'001 : max 8mA 3b'010 : max 12mA 3b'011 : max 16mA 3b'100 : max 24mA 3b'101 : max 32mA 3b'110 : max 46mA 3b'111 : max 64mA
6:4	SHORT_LEVEL	The SHORT_LEVEL determines the level at which a SHORT event occurs. 3b'000 : > 3V 3b'001 : > 4V 3b'010 : > 6V 3b'011 : > 8V 3b'100 : > 12V 3b'101 : > 16V 3b'110 : > 24V 3b'111 : > 36V
2:0	FB_LEVEL	The FB_LEVEL determines the level at which a FB event occurs. 3b'000 : < 0.40V 3b'001 : < 0.50V 3b'010 : < 0.60V 3b'011 : < 0.70V 3b'100 : < 0.85V 3b'101 : < 1.00V 3b'110 : < 1.15V 3b'111 : < 1.30V

FAULT CONTROL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x07	0x800	TIMEOUT_E	OVUV15_EN	x	MS_VS_LOCK	MS_VS_DIMM	MS_VS_DET_E	O_FB_E	O_DET_E	S_DET_E	T_OFF_E	S_OFF_E	O_OFF_E

Bits	Fields	Descriptions
11	TIMEOUT_E	If this bit is set, the internal timeout timer starts counting from receiving some read-back command and stops after receiving the first response bit from daisied devices. If the timer will be expired due to any no response, internal communication logics will be recovered to the standby state. The timeout counter is expired at approximately 819us upon receiving the read command. Recommended to be set.
10	OVUV15_EN	This enables the detection of OV15 and UV15 of the internal LDO output.
8	MS_VS_LOCK	Under the MS_VS_LOCK = 1 and the MS_VS_DET_E = 1, if the vsync missing event happens, then the dimming will be automatically fixed to PWM full value and the VREF FIX value until receiving a vsync.
7	MS_VS_DIMM	Under the MS_VS_DIMM = 1 and the MS_VS_DET_E = 1, if the vsync missing event happens, then the dimming out will be automatically fixed to the middle level brightness until receiving a vsync.
6	MS_VS_DET_E	This bit enables to detect whether the vsync input is missed or not. If the vsync is not supplied within about 26ms, then the vsync missing event is asserted internally. This event assertion can be readout through the FALUT STATUS register.
5	O_FB_E	This bit setting makes the FB possible to be asserted in only case the corresponding channel's OPEN event is not happen. If this bit is 0, then the FB can be asserted owing to the OPEN event.
4	O_DET_E	This bit setting enables to detect the OPEN event, and the event can be detected in only case the corresponding channel's PWM output is OFF status and its channel's headroom is below 0.25V.
3	S_DET_E	This bit setting enables to detect the SHORT event, and the event can be detected in only case the corresponding channel's PWM output is ON status and its channel's headroom is above the set value in the SHORT_LEVEL register.
2	T_OFF_E	This bit setting makes all channels' output be off in case the THERMAL event happens, and all channels will be activated again after the THERMAL event is cleared.
1	S_OFF_E	This bit setting makes the SHORTed channel's output be off in case the SHORT event happens.
0	O_OFF_E	This bit setting makes the OPENed channel's output be off in case the OPEN event happens.

MAX CURRENT VREF

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x08	0x000	MAX_CURR_VREF [11:0]											

Bits	Fields	Descriptions
11:0	MAX_CURR_VREF	The MAX_CURR_VREF determines the maximum current value of a frame to be dimmed. If the max current level is set to 8mA, then the MAX_CURR_VREF value is proportional to the max current level, and so this value decides the max current reference of all blocks being dimmed for a frame. The setting value will come into effect at the next vsync assertion even if the value is set during current dimming frame earlier than the next vsync.

DELAY CH EXTEND

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x09	0x000	DELAY_CH6 [13:12]		DELAY_CH5 [13:12]		DELAY_CH4 [13:12]		DELAY_CH3 [13:12]		DELAY_CH2 [13:12]		DELAY_CH1 [13:12]	

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0A	0x000	DELAY_CH12 [13:12]		DELAY_CH11 [13:12]		DELAY_CH10 [13:12]		DELAY_CH9 [13:12]		DELAY_CH8 [13:12]		DELAY_CH7 [13:12]	

Bits	Fields	Descriptions
11:0	DELAY_CHx [13:12]	This register includes the MSB 2 bits of each channel's DELAY_CHx [13:0] value. The DELAY_CHx determines from which PWM position the dimming out starts. If this is set to n, then the dimming out starts at $(n+1)^{th}$ PWM position. Normally, the setting value can not exceed the max PWM resolution.

DELAY CH1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0B	0x000	DELAY_CH1 [11:0]											

DELAY CH2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0C	0x000	DELAY_CH2 [11:0]											

DELAY CH3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0D	0x000	DELAY_CH3 [11:0]											

DELAY CH4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0E	0x000	DELAY_CH4 [11:0]											

DELAY CH5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0F	0x000	DELAY_CH5 [11:0]											

DELAY CH6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x10	0x000	DELAY_CH6 [11:0]											

DELAY CH7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x11	0x000	DELAY_CH7 [11:0]											

DELAY CH8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x12	0x000	DELAY_CH8 [11:0]											

DELAY CH9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x13	0x000	DELAY_CH9 [11:0]											

DELAY CH10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x14	0x000	DELAY_CH10 [11:0]											

DELAY CH11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x15	0x000	DELAY_CH11 [11:0]											

DELAY CH12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x16	0x000	DELAY_CH12 [11:0]											

SERIALIZER OUTPUT BAUDRATE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x25	0x318	SERIAL_CLK_LOW [5:0]						SERIAL_CLK_HIGH [5:0]					

Bits	Fields	Descriptions
11:6	SERIAL_CLK_LOW	These bits determine the number of MCLK of V_{DD} level in logic low value ‘0’ or the number of MCLK of 0 level in logic high value ‘1’ when the XD12 is driving the D_{IO} pin on read-back. For more details, refer to Table. 4, Logic low time.
5:0	SERIAL_CLK_HIGH	These bits determine the number of MCLK of V_{DD} level in logic high value ‘1’ or the number of MCLK of 0 level in logic low value ‘0’ when the XD12 is driving the D_{IO} pin on read-back. For more details, refer to Table. 4, Logic high time.

SERIAL LATENCY

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x26	0x078	x	x	x	x	x	SERIAL_LATENCY [6:0]						

Bits	Fields	Descriptions
6:0	SERIAL_LAT	This determines the interval time between the successive data packets incoming from the XC. That means the interval time between the last bit of n^{th} XD device’s data packet and the first bit of $(n+1)^{th}$ XD device’s data packet incoming from the XC. The setting value should be larger than the 1-bit serial data length time at which the XC is sending to the daisied XDs. So, the proper value depends on the serial baudrate of the XC and can be set between (1-bit time + 3 MCLKs) and (1-bit time + 5 MCLKs). If the proper value will be set, then the serial data value will be broken.

MCLK LOCK

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x27	0xFB0	MCLK_LOCK_CNT [11:0]											

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x28	0x84F	MCLK_LOCK_E	x	FLL_RANGE [1:0]		MCLK_LOCK_CNT [19:12]							

Bits	Fields	Descriptions
11:0 7:0	MCLK_LOCK_CNT	This determines the FLL target frequency value. The proper value according to the frame frequency is showed in the Table .
11	MCLK_LOCK_E	This bit enables the FLL function. The FLL continues to calibrate the MCLK frequency to exactly match the PWM frequency with the time of one frame. 0 Enable FLL function (= The MCLK is unlocking, so calibrated to the target frequency) 1 Disable FLL function (= The MCLK is locking at the current frequency)
9:8	FLL_RANGE	It is recommended to use the default value.

TEMP

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x29	0x820	OFS_TEMP]3:0]				x	DAC_RNG	FLT_CTL [1:0]		x	x	FLT_GAIN [1:0]	

Bits	Fields	Descriptions
11:8	OFS_TEMP	It is recommended to use the default value.
6	DAC_RNG	It is recommended to use the default value.
5:4	FLT_CTL	It is recommended to use the default value.
1:0	FLT_GAIN	It is recommended to use the default value.

OSC FLL MANUAL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x2A	0x000	OSC_FLL_MAN [11:0]											

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x2B	0x808	OSC_MAN_E								OSC_FLL_MAN [15:12]			

Bits	Fields	Descriptions
11	OSC_MAN_E	Enable the manual mode of the FLL module.
11:0 3:0	OSC_FLL_MAN	This determines the FLL target frequency value in manual mode.

OSC FLL MONITOR

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
RO	0x2C	0x000	x	x	x	x	x	OSC_FLL_FLT [6:0]						

Bits	Fields	Descriptions
6:0	OSC_FLL_FLT	

VREF FIX

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x2F	0x000	VREF_FIX [11:0]											

Bits	Fields	Descriptions
11:0	VREF_FIX	It determines the fixed VREF out value instead of the MAX_CURR_VREF when the vsync miss is detected under the MISS_VS_DIMM=1 and the MISS_VS_DET_E=1. At the same time, the LD PWM data is out with full value.

These registers can not go outside the Global Technologies.

OP MODE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3F	0x000	TEST_EN	DDIO_DIS	TEST_ANA_EN[1:0]		x	PWM_FULL_O	MCLK32_O	VREF_O	x	x	x	ADDR_EXT

Bits	Fields		Descriptions
11	TEST_EN	0	NOP
		1	It is used to test the device for trimming. To use the PWM_FULL_O, MCLK_O and VREF_O function, this bit should be set firstly and cleared lastly after these functions are used.
10	DDIO_DIS	0	NOP
		1	Under TEST_EN=1, the D_{DIO} pin's output will be disabled so that the read access of registers is possible without any mulfunction even though the MCLK32 clock is driven on the D_{DIO} pin.
9:8	TEST_ANA_EN		
6	PWM_FULL_O	0	NOP
		1	Under TEST_EN =1, the enabled channels' PWM out keeps active state regardless of the vsync.
5	MCLK32_O	0	NOP
		1	Under TEST_EN =1, the clock obtained by dividing MCLK by 32 is output to DDIO pin.
4	VREF_O	0	NOP
		1	Under TEST_EN =1, the MAX_CURR_VREF register value is continuously output to all internal current control logics regardless of the vsync.
0	ADDR_EXT		This bit is used to select which register groups, the general registers or the OTP mirror registers, will be accessed. The OTP control registers address, 0x3A ~ 0x3F, can be commonly accessed regardless of this bit.
		0	The general registers can be accessed except for the OTP mirror registers.
		1	The OTP mirror registers can be accessed except for the general registers.

OTP ACCESS

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3A	0x000	x	x	x	x	x	x	x	x	OTP_PG_ACCESS_CYCLE [15:12]			

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3B	0x3FF	OTP_PG_ACCESS_CYCLE [11:0]											

OTP WRITE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3C	0x004	x	x	x	x	x	x	x	x	OTP_WSEL [3:0]			

OTP RD/PROG

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3D	0x000	x	x	x	x	x	x	x	x	x	x	OTP_RD_START	OTP_PG_START

OTP PROTECT

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3E	0x5A5	PROTECT_EN/DIS [11:0] ※ enable (0x5A5), disable (0xA5A), need to disable for writing											

These registers can not go outside the Global Technologies.

These registers are the OTP Mirror registers.

OTP CRC

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x00	0x000	x	x	x	x	OTP_CRC_CHECKSUM [7:0]							

OSC

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x01	0x020	x	x	x	x	x	x	OSC [5:0]					

VREF CTL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x02	0x020	x	x	x	x	x	x	VREF_CTL [5:0]					

These registers can not go outside the Global Technologies.

ICTL_L_CH 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x03	0x40	x	x	x	x	x	ICTL_L_CH1 [6:0]						

ICTL_L_CH 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x04	0x40	x	x	x	x	x	ICTL_L_CH2 [6:0]						

ICTL_L_CH 3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x05	0x40	x	x	x	x	x	ICTL_L_CH3 [6:0]						

ICTL_L_CH 4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x06	0x40	x	x	x	x	x	ICTL_L_CH4 [6:0]						

ICTL_L_CH 5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x07	0x40	x	x	x	x	x	ICTL_L_CH5 [6:0]						

ICTL_L_CH 6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x08	0x40	x	x	x	x	x	ICTL_L_CH6 [6:0]						

ICTL_L_CH 7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x09	0x40	x	x	x	x	x	ICTL_L_CH7 [6:0]						

ICTL_L_CH 8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0A	0x40	x	x	x	x	x	ICTL_L_CH8 [6:0]						

ICTL_L_CH 9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0B	0x40	x	x	x	x	x	ICTL_L_CH9 [6:0]						

ICTL_L_CH 10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0C	0x40	x	x	x	x	x	ICTL_L_CH10 [6:0]						

ICTL_L_CH 11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0D	0x40	x	x	x	x	x	ICTL_L_CH11 [6:0]						

ICTL_L_CH 12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0E	0x40	x	x	x	x	x	ICTL_L_CH12 [6:0]						

These registers can not go outside the Global Technologies.

ICTL_H_CH 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1B	0x40	x	x	x	x	x	ICTL_H_CH1 [6:0]						

ICTL_H_CH 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1C	0x40	x	x	x	x	x	ICTL_H_CH2 [6:0]						

ICTL_H_CH 3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1D	0x40	x	x	x	x	x	ICTL_H_CH3 [6:0]						

ICTL_H_CH 4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1E	0x40	x	x	x	x	x	ICTL_H_CH4 [6:0]						

ICTL_H_CH 5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1F	0x40	x	x	x	x	x	ICTL_H_CH5 [6:0]						

ICTL_H_CH 6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x20	0x40	x	x	x	x	x	ICTL_H_CH6 [6:0]						

ICTL_H_CH 7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x21	0x40	x	x	x	x	x	ICTL_H_CH7 [6:0]						

ICTL_H_CH 8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x22	0x40	x	x	x	x	x	ICTL_H_CH8 [6:0]						

ICTL_H_CH 9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x23	0x40	x	x	x	x	x	ICTL_H_CH9 [6:0]						

ICTL_H_CH 10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x24	0x40	x	x	x	x	x	ICTL_H_CH10 [6:0]						

ICTL_H_CH 11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x25	0x40	x	x	x	x	x	ICTL_H_CH11 [6:0]						

ICTL_H_CH 12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x26	0x40	x	x	x	x	x	ICTL_H_CH12 [6:0]						

Soldering Temperature Reflow Profile

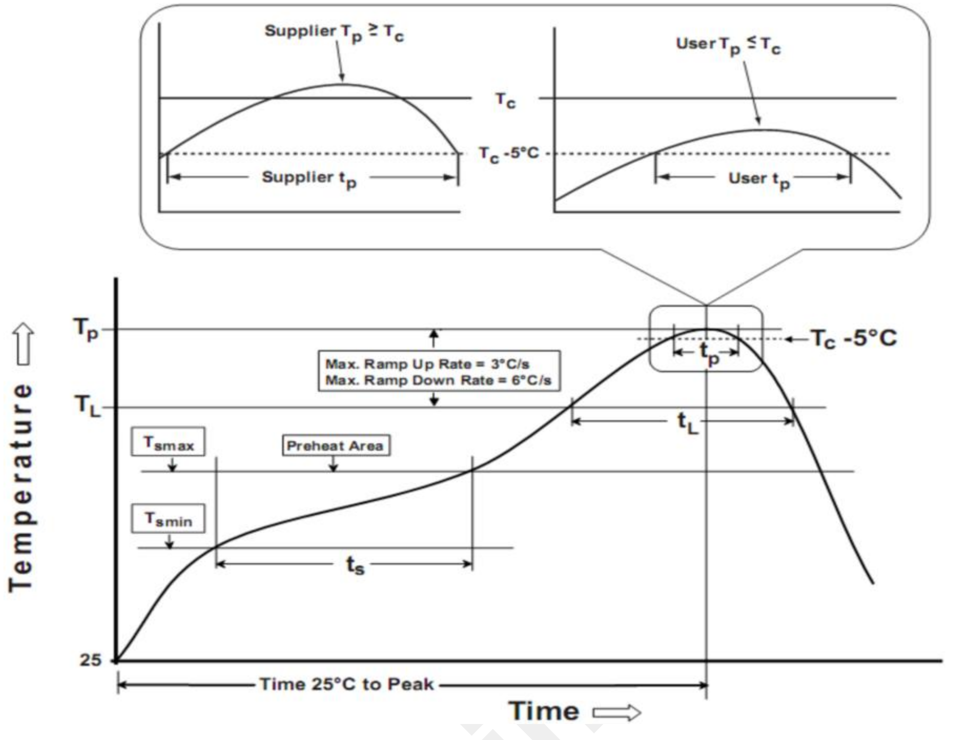


Figure 15. Reflow Soldering information

Parameter	Pb-free soldering conditions
Average ramp-up rate (217 ℃ to Peak)	3℃/second max.
Time of Preheat temp (from 150 ℃ to 200 ℃)	60 ~ 120 second
Time to be maintained above 217 ℃	60 ~ 150 second
Peak temperature	260 ℃
Time within 5 ℃ of actual peak temp	30 second
Ramp-down rate	6 ℃/second max.
Time from 25℃ to peak temp	8 minutes max.

Table 8. Reflow Soldering Parameter

Note:

1. The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “ Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”
2. Tolerance for peak profile temperature (Tp) is defined as a supplier minimum and a user maximum.
3. This document should serve as recommendation only. Other parameters may also affect soldering, so these profiles do not guarantee absolute success.
4. Soldering profile should be determined by the manufacturer of the solder paste, providing there is no contradiction with the recommendations in this document.

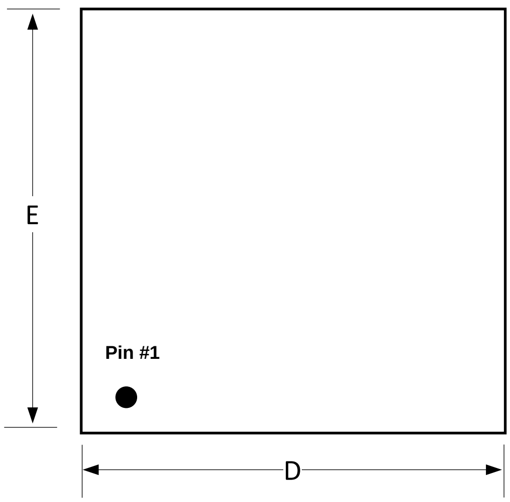
Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

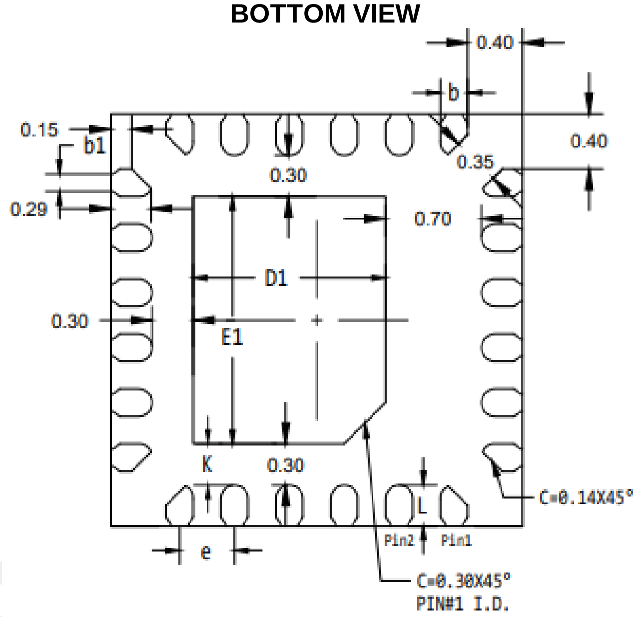
Package specification

QFN 3030-24L Package Outline

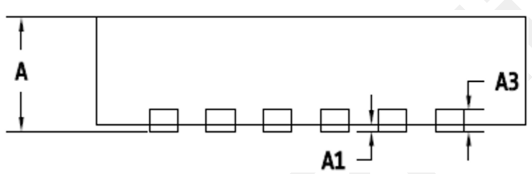
TOP VIEW



BOTTOM VIEW



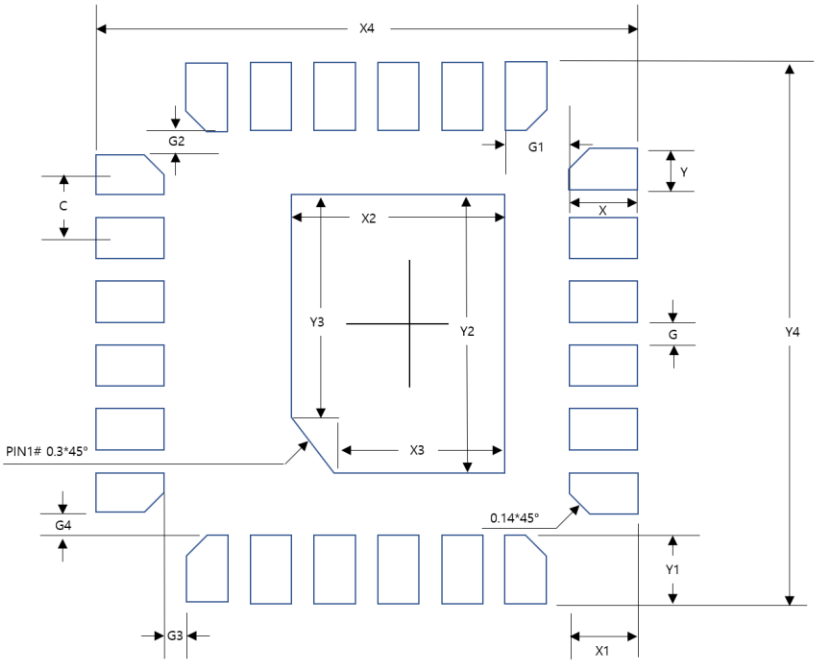
SIDE VIEW



Symbol	Dimensions in mm		
	Min	Typ	Max
A	0.50	0.55	0.60
A1	0.00	-	0.05
A3	0.152 REF		
b	0.15	0.20	0.25
b1	0.08	0.13	0.18
D	2.90	3.00	3.10
D1	1.25	1.30	1.35
E	2.90	3.00	3.10
E1	1.75	1.80	1.85
e	0.40 BSC		
K	0.25	0.30	0.35
L	0.25	0.30	0.35

Figure 16. Package Outline

Recommended Footprint



Symbol	Dimensions in mm	
	Value	Margin (Typ+)
X	0.5	+0.2
Y	0.2	0
C	0.4	0
X1	0.5	+0.2
Y1	0.5	+0.2
X2	1.4	+0.1
Y2	1.9	+0.1
X3	1.1	+0.1
Y3	1.6	+0.1
X4	3.4	+0.2
Y4	3.4	+0.2
G	0.2	0
G1	0.3	0
G2	0.1	0
G3	0.1	0
G4	0.1	0

Figure 17. Recommended Footprint

Note: The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”